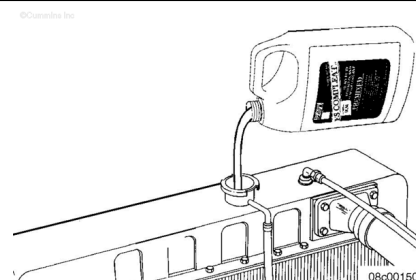


Trainings ▾

Fully Formulated Coolant/Antifreeze

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Note : Low-silicate antifreeze meeting ASTM D4985 is inadequate for extended service intervals of 80,467 to 241,402 km [50,000 to 150,000 mi].

Cummins Inc. recommends using either a 50/50 mixture of high-quality water and fully formulated antifreeze, or fully formulated coolant when filling the cooling system. The fully formulated antifreeze or coolant **must** meet Cummins® Engineering Standard (CES) 14603 specifications.

Most coolants which meet ASTM D6210 also meet CES 14603. However, some OAT coolants such as Shell™ Rotella™ ELC, Chevron™ Delo™ ELC and their private label counterparts meet ASTM D6210 but do **not** meet the elastomer compatibility test of CES 14603. These coolants are acceptable for use, assuming the OEM added silicate at initial fill. Use Section 3 - Extended Service Interval of the following bulletin for more details. Refer to Cummins® Coolant Requirements and Maintenance, Bulletin 3666132. (</qs3/pubsys2/xml/en/bulletin/3666132.html>)

High-quality water is important for cooling system performance. Excessive levels of calcium and magnesium contribute to scaling problems, and excessive levels of chlorides and sulfates cause cooling system corrosion.

Water Quality	
Calcium Magnesium (Hardness)	Maximum 170 ppm as (CaCO ₃ + MgCO ₃)
Chloride	Maximum 40 ppm as (Cl)
Sulfate	Maximum 100 ppm as (SO ₄)

Cummins Inc. recommends the use of Fleetguard® Compleat ES™. It is fully formulated and meets or exceeds the following standards:

- CES 14603
- ASTM D6210
- TMC RP329
- TMC RP330.

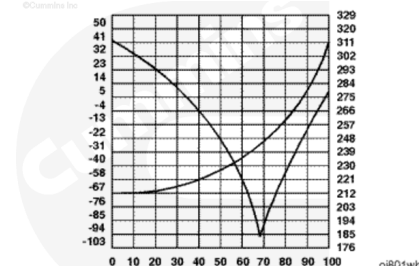
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Fully formulated antifreeze **must** be mixed with quality water at a 50/50 ratio (40-percent to 60-percent working range). A 50/50 mixture of antifreeze and water gives a -36°C [-33°F] freezing point and a 110°C [230°F] boiling point, which is adequate for locations in North America. The actual lowest freezing point of ethylene glycol antifreeze occurs at 68 percent. The use of higher concentrations of antifreeze raises the freezing point of the solution and increases the possibility of a silica gel problem.

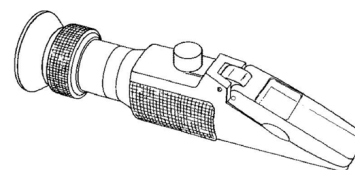
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A refractometer **must** be used to measure the freezing point of the coolant accurately.

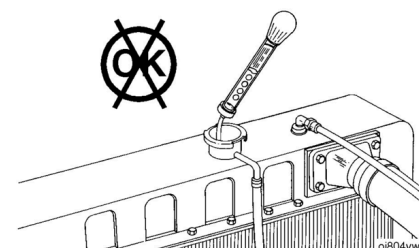
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ra8toda

Do **not** use a floating ball hydrometer. The use of a floating ball hydrometer can give an incorrect reading.

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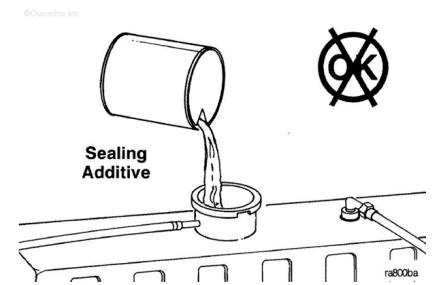
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Cooling System Sealing Additives

Do **not** use sealing additives in the cooling systems. The use of sealing additives can:

- Build up in coolant low-flow areas

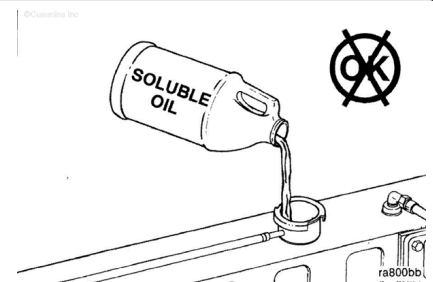
- Clog coolant filters
- Plug radiators and oil coolers.



Cooling System Soluble Oils

Do **not** use soluble oils in the cooling system. The use of soluble oils can:

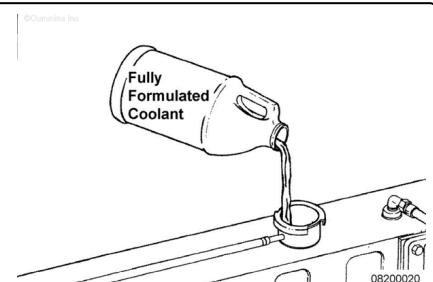
- Allow cylinder liner pitting
- Corrode brass, aluminum, and copper
- Damage heat transfer surfaces
- Damage seals and hoses.



Supplemental Coolant Additive (SCA)

⚠ CAUTION ⚠

Signature™, ISX, and QSX15 engines use aluminum parts that are in contact with the coolant. Improper coolant, coolant filter selection, and maintenance will likely result in corrosion of any of these parts. Insufficient concentration of the coolant additives will result in liner pitting and engine failure.



Fully formulated products contain SCAs and are required to protect the cooling system from scale and fouling, solder corrosion, and general corrosion. The coolant filter is required to protect the cooling system from abrasive materials, debris, and precipitated coolant additives.

Fleetguard® DCA4 Service Filters and Liquid Precharge

Note : Filters must meet Cummins Inc. Source Approval Method 10,769. Fleetguard® filters meet Source Approval Method 10769. The standard filter for the Signature™ engine is Fleetguard® WF2126. For systems larger than 100 gallons, use 1 quart per 15 gallons. Consult vehicle manufacturer for total cooling system capacity.

WF2126 - This filter is designed for service intervals up to 80,467 km [50,000 mi] when using TMC RP329/RP330 or ASTM D6210 coolants that meet Cummins® Engineering Standard CES 14603. The filter is designed for cooling systems up to 49 liters [13 gal]. This filter contains 8 units of DCA4.

WF2127 - This filter has been designed for extended service intervals of 80,467 to 241,402 km [50,000 to 150,000 mi]. It has no chemical additives, as it contains zero units of DCA4. This filter can be used in the following systems:

- Cooling systems of any capacity
- See maintenance chart below.

When using the WF2127 filter, the following volumes of liquid treatment listed in the table **must** be added at the designated mileages/intervals.

Cooling System Capacity	80,467 km [50,000 mi] Service Interval with RP329/330 Coolant (DCA2 or DCA4)	241,402 km [150,000 mi] Service Interval with ES Compleat™ Coolant (ES Extender Liquid)
0 to 15 gal	1 qt (10 units)	1 qt (15 units)
16 to 30 gal	2 qt (20 units)	2 qt (30 units)
31 to 45 gal	3 qt (30 units)	3 qt (45 units)
46 to 60 gal	4 qt (40 units)	4 qt (60 units)
61 to 75 gal	5 qt (50 units)	5 qt (75 units)
76 to 90 gal	6 qt (60 units)	6 qt (90 units)
91 to 150 gal	7 qt (70 units)	7 qt (105 units)

Testing SCA Concentration Level CC-2602 Test Kit

Precautions and Instructions for Proper Kit Use

Carefully follow the instructions to test the coolant. Take the appropriate action recommended by the kit.

- The temperature of the coolant sample to be tested **must** be between 10°C [50°F] and 54°C [130°F] to obtain correct results.
- To get the best color match results, compare test strip pads to the color chart in daylight or under cool white fluorescent lighting. If unsure about a specific color match when a test falls

between two colors on the color chart, choose the lower numbered (or lettered) block. It is safer to underestimate the results than to overestimate.

- The test strips do have a limited shelf life and are sensitive to humidity and extreme heat. Proper handling and storage is necessary to protect the life of the strips.
- Keep the cap tightly sealed on the test strip bottle **except** when removing a strip. Store away from direct sunlight and in an area where the temperature will generally stay below 32°C [90°F].
- Do **not** use the test strips after the expiration date stamped on the bottle.
- Discard the kit if the top pads on the unused strips have turned light brown.
- Use one strip at a time and take care **not** to touch any of the pads on the strip. This will contaminate the pads and skew the test results.
- If the strip container is left uncapped for 24 hours, moisture in the air will render the strips useless, even though no discoloration will be evident.
- **Only** use the color chart supplied with the kit.
- Following the correct test times is very important. Use a clock or stopwatch.
- Do **not** use the test kit to maintain minimum SCA concentration levels (i.e., 1.5 units).
- When performing service that requires draining the cooling system, take special precautions to collect coolant in a clean non-galvanized container, seal to prevent contamination, and save for reuse.

Coolant Testing

Probablizer:

- 3318169S Plug - Installs on the engine for easy coolant sampling.
- 3318168S Cap - Use with Monitor C™ bottle to sample coolant.
- CC2700 Monitor C™ - Use lab analysis of coolant samples for more detailed analysis.

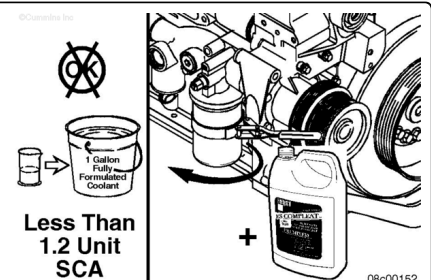
CC2602 Coolant Test Kit

- Works with any SCA formulation. Call 1-800-22FILTER (800-223-4583) if using this test kit and the color chart does **not** show the number of units of SCA per gallon of coolant.

Test Intervals

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Testing is recommended if the operator is **not** sure of the cooling system condition due to leaks, uncontrolled topping off of the system, or major coolant loss.

If the SCA concentration is below 1.2 units per gallon, replace the filter and precharge with liquid.

Check the SCA concentration level at least every 6 months, any time the coolant condition is unknown, or if corrosion is apparent within the cooling system.

Use Fleetguard® coolant test kit, CC2602, to check the concentration level. Instructions are included with the test kit.

Call the following numbers to get answers to any questions regarding cooling system maintenance:

Cummins®	Fleetguard®
1-800-CUMMINS or visit www.cummins.com/customer-assistance	1-800-22FILTER
1-800-266-6467	1-800-223-4583

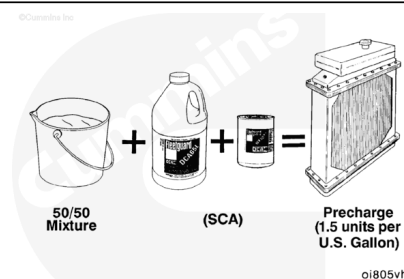
Coolant Replacement Requirements

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Drain and flush the cooling system after 6000 hours or 3 years of service. However, if Fleetguard® ES coolant and ES filters are used, check chloride, sulfate, and pH levels according to Cummins® Coolant Requirements and Maintenance, Bulletin

3666132 (/qs3/pubsys2/xml/en/bulletin/3666132.html), to determine whether the coolant **must** be replaced. Fill with either new fully formulated coolant or ES coolant.

Use the following bulletin for additional information on coolant and coolant system recommendations. Refer to Cummins® Coolant Requirements and Maintenance, Bulletin 3666132. (/qs3/pubsys2/xml/en/bulletin/3666132.html)

Last Modified: 24-Sep-2018
